

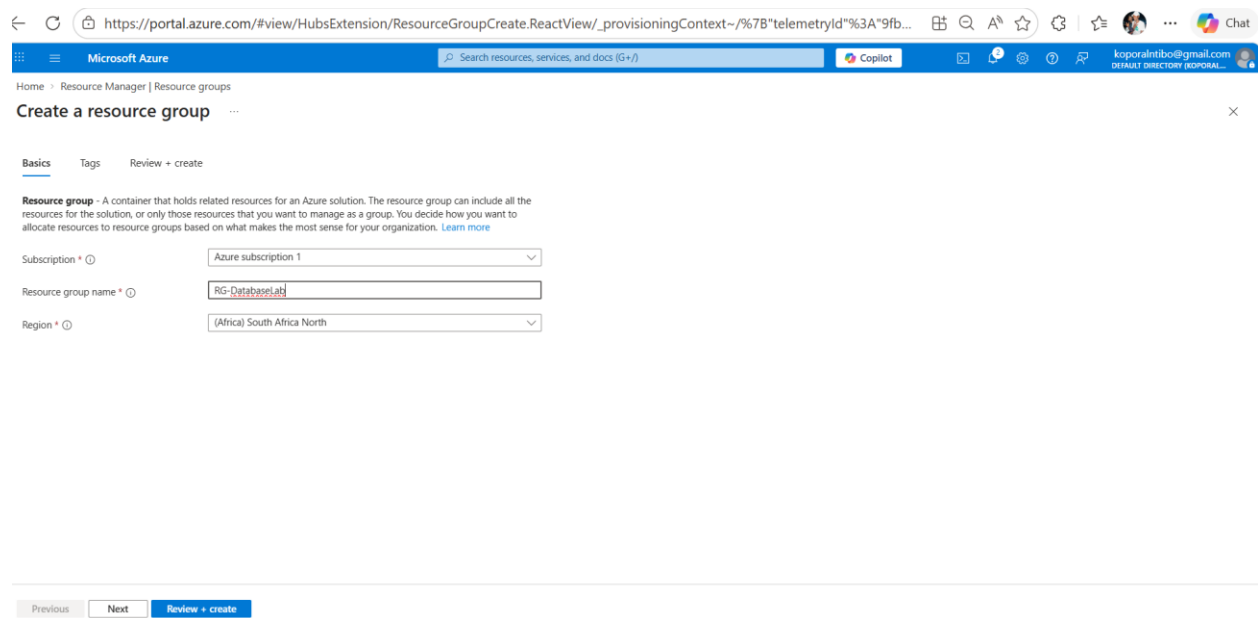
CHAPTER 05 LAB

AZURE DATABASE SERVICES

LAB 5: Implementing and Managing Azure Databases

TASK 1: CREATE AZURE SQL DATABASE

I chose Azure subscription 1 for this lab because it provides access to the necessary Azure services under my student account, allowing me to create and manage resources such as virtual machines and databases without additional cost. This subscription supports the deployment of cloud infrastructure required for the exercise, making it suitable for building and testing solutions within the Azure environment.



The screenshot shows the Azure portal interface for creating a new resource group. The browser address bar displays the URL: `https://portal.azure.com/#view/HubsExtension/ResourceGroupCreate.ReactView/_provisioningContext~/7B%20telemetryId%3A%209fb...`. The page title is "Create a resource group". Below the title, there are tabs for "Basics", "Tags", and "Review + create". The "Basics" tab is active. A description of a resource group is provided: "Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)". The form contains three fields: "Subscription" with a dropdown menu showing "Azure subscription 1", "Resource group name" with a text input field containing "RG-DatabaseLab", and "Region" with a dropdown menu showing "(Africa) South Africa North". At the bottom of the page, there are three buttons: "Previous", "Next", and "Review + create".

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name * ✓

Server * ⓘ ✓

[Create new](#)

Want to use SQL elastic pool? ⓘ ☐ Yes ☒ No

Workload environment ☐ Development

☒ Production

i Default settings provided for Production workloads. Configurations can be modified as needed.

Compute + storage * ⓘ

Hyperscale

Standard-series (Gen5), 2 vCores, zone redundant disabled

[Configure database](#)

Microsoft Azure

Home > Create a resource > Marketplace

Create SQL Database Server

Microsoft

Server name * ✓
 .database.windows.net

Location * ✓

Authentication

i Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication. [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin. [Learn more](#)

Authentication method

- ☐ Use Microsoft Entra-only authentication
- ☒ Use both SQL and Microsoft Entra authentication
- ☐ Use SQL authentication

Set Microsoft Entra admin

[korporalntibo@gmail.com#EXT#@korporalntibogmail.onmicrosoft.com](#)
Admin Object/App ID: 7a03aa2d-d4a1-4d61-91d8-cec55bc2d02e
[Set admin](#)

Server admin login * ✓

Password * ✓

[OK](#)

[Feedback](#)

Microsoft Azure

Home > Create a resource > Marketplace

Create SQL Database

Microsoft

creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method *

☐ No access

☒ Public endpoint

☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to 'Yes' allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to 'Yes' will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

Add current client IP address *

Connection policy

Configure how clients communicate with your SQL Database server. [Learn more](#)

Connection policy

☒ Default - Uses Redirect policy for all client connections originating inside of Azure (except Private Endpoint connections) and Proxy for all client connections originating outside Azure

☐ Proxy - All connections are proxied via the Azure SQL Database gateways

☐ Redirect - Clients establish connections directly to the node hosting the database

Cost summary

Hyperscale (HS_Gen5_2) - Primary replica	
Cost per vCore (in USD)	178.68
vCores selected	x 2
Hyperscale (HS_Gen5_2) - HA replicas	
Cost per vCore (in USD)	178.68
vCores selected	x 2
HA replicas	x 0
ESTIMATED COMPUTE COST / MONTH	357.36 USD
STORAGE COST / GB / MONTH	0.33 USD

STORAGE NOTES

¹ In the Hyperscale tier, storage costs are calculated based on actual allocation. Allocated space increases automatically as needed, up to 100 TB.

Microsoft Azure

Home > Create a resource > Marketplace

Create SQL Database

Microsoft

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription	Azure subscription 1
Resource group	RG-DatabaseLab
Region	South Africa North
Database name	StudentDatabase
Server	(new) sqlserver-siyabonga
Authentication method	SQL and Microsoft Entra authentication
Server admin login	dbadmin
Microsoft Entra Admin	koporalntibo@gmail.com#EXT#@koporalntibogmail.onmicrosoft.com
Compute + storage	Hyperscale: Standard-series (Gen5), 2 vCores, zone redundant disabled
Backup storage redundancy	Geo-redundant backup storage

Networking

Allow Azure services and resources to access this server	Yes
Add current client IP address	Yes
105.244.78.238	
Private endpoint	None

Cost summary

Hyperscale (HS_Gen5_2) - Primary replica	
Cost per vCore (in USD)	178.68
vCores selected	x 2
Hyperscale (HS_Gen5_2) - HA replicas	
Cost per vCore (in USD)	178.68
vCores selected	x 2
HA replicas	x 0
ESTIMATED COMPUTE COST / MONTH	357.36 USD
STORAGE COST / GB / MONTH	0.33 USD

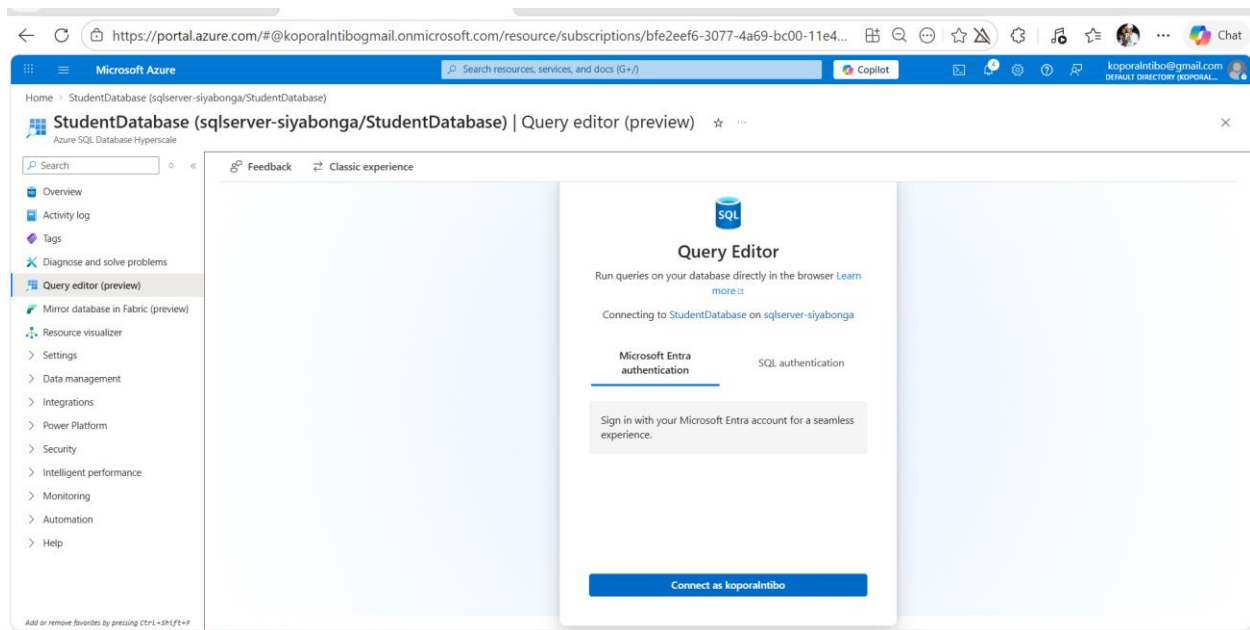
STORAGE NOTES

¹ In the Hyperscale tier, storage costs are calculated based on actual allocation. Allocated space increases automatically as needed, up to 100 TB.

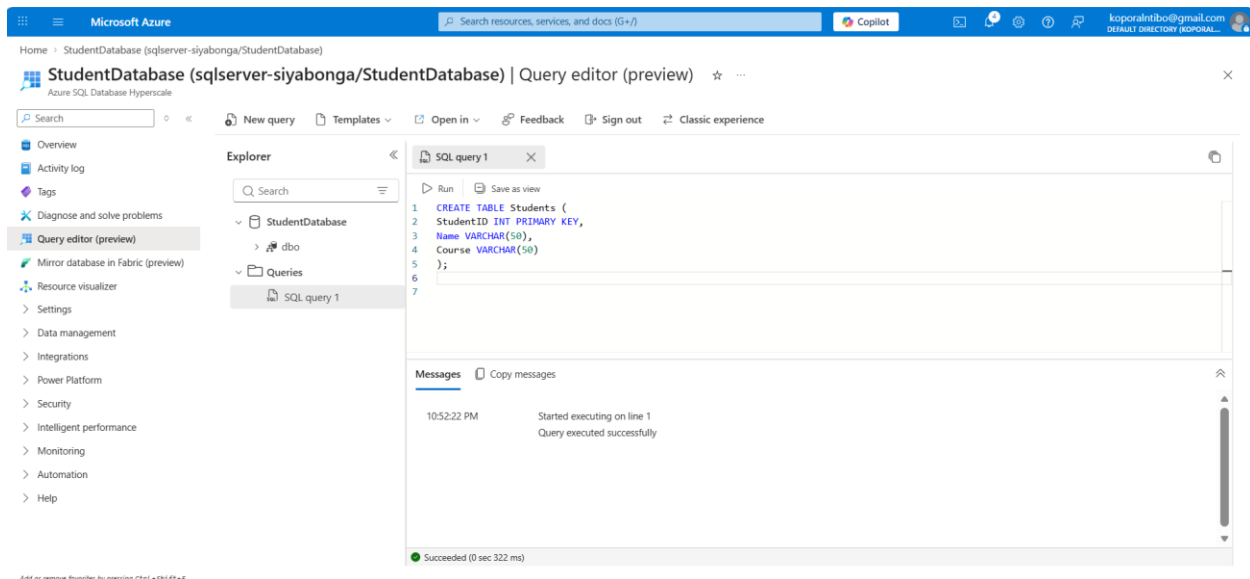
[Download a template for automation](#)

Task 2: Create Database Table

After the server is created and deployed, enter the Query Editor tab and select the SQL authentication option. Once the SQL authentication option is selected, use the server authentication credentials to connect to the server.



Once connected to the server, enter the SQL commands provided to create a SQL table structure in the server, then run it.



Task 3: Insert Data into Table

← ↻ 🔍 <https://portal.azure.com/#@korporalntibogmail.onmicrosoft.com/resource/subscriptions/bfe2eef6-3077-4a69-bc00-11e4...> Copilot korporalntibogmail.com

Home > StudentDatabase (sqlserver-siyabonga/StudentDatabase) | Query editor (preview) ☆

Search New query Templates Open in Feedback Sign out Classic experience

Overview Activity log Tags Diagnose and solve problems Query editor (preview) Mirror database in Fabric (preview) Resource visualizer Settings Data management Integrations Power Platform Security Intelligent performance Monitoring Automation Help

Explorer

- StudentDatabase
 - dbo
 - Queries
 - SQL query 1

SQL query 1

```
1 INSERT INTO Students VALUES (1,'Alice','Cloud Computing');
2 INSERT INTO Students VALUES (2,'Bob','Cyber Security');
3 INSERT INTO Students VALUES (3,'Thabo','Networking');
```

Messages Copy messages

10:56:00 PM Started executing on line 1
Query executed successfully

Succeeded (0 sec 237 ms)

← ↻ 🔍 <https://portal.azure.com/#@korporalntibogmail.onmicrosoft.com/resource/subscriptions/bfe2eef6-3077-4a69-bc00-11e4...> Copilot korporalntibogmail.com

Home > StudentDatabase (sqlserver-siyabonga/StudentDatabase) | Query editor (preview) ☆

Search New query Templates Open in Feedback Sign out Classic experience

Overview Activity log Tags Diagnose and solve problems Query editor (preview) Mirror database in Fabric (preview) Resource visualizer Settings Data management Integrations Power Platform Security Intelligent performance Monitoring Automation Help

Explorer

- StudentDatabase
 - dbo
 - Tables
 - Students
 - Views
 - Stored Procedures
 - Functions
 - Queries
 - SQL query 1

SQL query 1

Data preview - Students Showing 1000 rows Search

	ID	StudentID	ABC Name	ABC Course
1	1		Alice	Cloud Computing
2	2		Bob	Cyber Security
3	3		Thabo	Networking

Succeeded (0 sec 185 ms) Columns: 3 Rows: 3

Task 4: Query Data from Database

Use `SELECT * FROM Students` to display the full table with its' respective data already filled in.

The screenshot shows the Azure portal interface for the 'StudentDatabase (sqlserver-siyabonga/StudentDatabase)'. The 'Query editor (preview)' is active, displaying a SQL query: `SELECT * FROM Students;`. The query has been executed successfully, returning 3 rows of data. The results are displayed in a table with the following columns: **StudentID**, **Name**, and **Course**.

StudentID	Name	Course
1	Alice	Cloud Computing
2	Bob	Cyber Security
3	Thabo	Networking

The status bar at the bottom indicates 'Succeeded (0 sec 139 ms)' and 'Columns: 3 Rows: 3'.

Task 5: Create Cosmo DB (NoSQL Database)

Once in the “Create Azure Cosmos DB Account – Azure Cosmos DB for NoSQL” framework, choose the learning workload as it corresponds with the subscription you have chosen which will be automatically selected, along with the resource group, due to it associated with the resource group you created before. Learning workload is best for beginners as it offers low cost, easy setup, and quick exploration to learn Azure Cosmos DB. Name the Cosmos DB account using your name to distinguish it from other potential accounts. Afterwards create the database.

The screenshot shows the 'Create Azure Cosmos DB Account - Azure Cosmos DB for NoSQL' form. The form is filled out with the following details:

- Project Details:**
 - Workload Type:** Learning (Best for beginners. Low cost, easy setup, and quick exploration to learn Azure Cosmos DB.)
- Subscription:** Azure subscription 1
- Resource Group:** RG-DatabaseLab (with a 'Create new' link)
- Instance Details:**
 - Account Name:** cosmos-siyabonga (with a checkmark indicating it's valid)
 - Availability Zones:** ☒ Enable ☐ Disable
 - Location:** Loading...

At the bottom, there are buttons for 'Review + create', 'Previous', and 'Next: Global distribution', along with a 'Feedback' link.

← ↻ <https://portal.azure.com/#create/Microsoft.DocumentDB> Microsoft Azure Search resources, services, and docs (G+)

Home

Create Azure Cosmos DB Account - Azure Cosmos DB for NoSQL

Validation failed. Required information is missing or not valid.

Azure Cosmos DB is a fully managed NoSQL and relational database service for building scalable, high performance applications. Go to production starting at \$24/month per database, multiple containers included. [Learn more](#)

Project Details

Choose a workload type that best aligns with your goals. This helps us provide an optimized starting point for your Azure Cosmos DB account setup. Don't worry—no matter which workload type you choose, you can customize each setting to fit your needs or stick to the defaults provided.

Workload Type *
Best for beginners. Low cost, easy setup, and quick exploration to learn Azure Cosmos DB.

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *
Resource Group *
[Create new](#)

Instance Details

Account Name * ✓

Configure availability zone settings for your account. You cannot change these settings once the account is created.

Availability Zones ☒ Enable ☐ Disable

Location *
Available locations are determined by your subscription's region and availability zones (if that is enabled). If you don't see a region select your desired location, please open a support request for more options.

[Review + create](#) [Previous](#) [Next: Global distribution](#) [Feedback](#)

← ↻ <https://portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2Fbfe2eef6-3077-...> Microsoft Azure Search resources, services, and docs (G+)

Home

Microsoft.Azure.CosmosDB-20260402231019 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

Inputs Outputs Template

Your deployment is complete

Deployment name : Microsoft.Azure.CosmosDB-20260402231019 Start time : 4/2/2026, 11:10:25 PM
Subscription : Azure subscription 1 Correlation ID : 7dc21cae-c14e-4602-9a0d-8343d660d940
Resource group : RG-DatabaseLab

Deployment details

Resource	Type	Status	Operation details
cosmos-siyabonga	Microsoft.DocumentDb/databases	OK	Operation details

Next steps

[Go to resource](#)

Give feedback

[Tell us about your experience with deployment](#)

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill. [Set up cost alerts >](#)

Microsoft Defender for Cloud

Secure your apps and infrastructure [Go to Microsoft Defender for Cloud >](#)

Free Microsoft tutorials

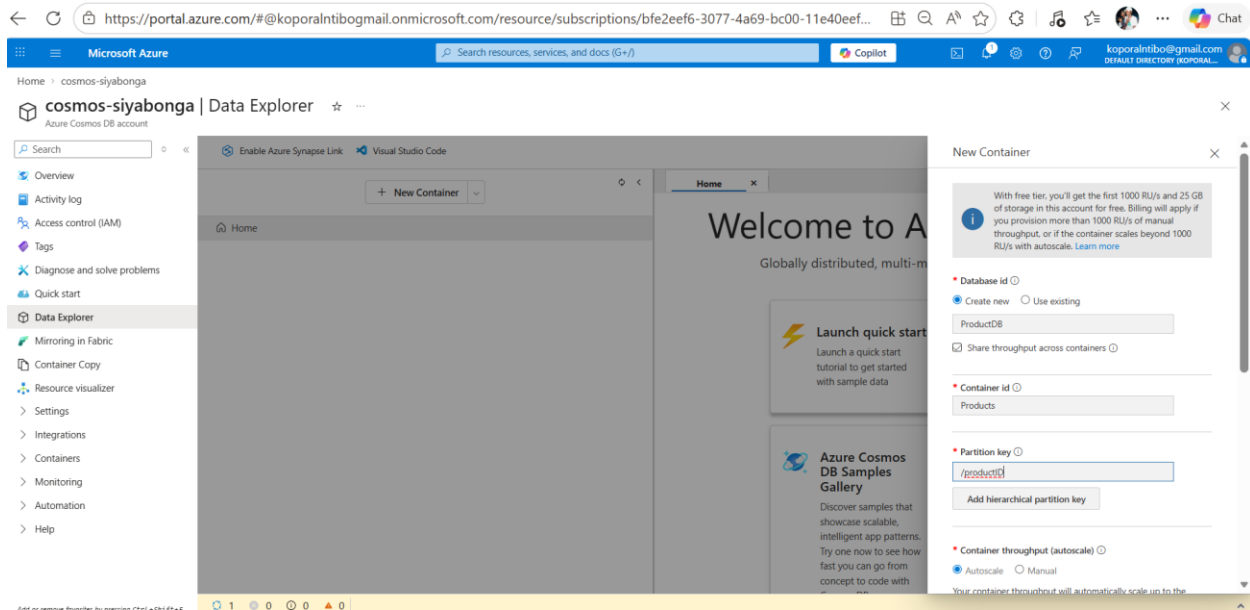
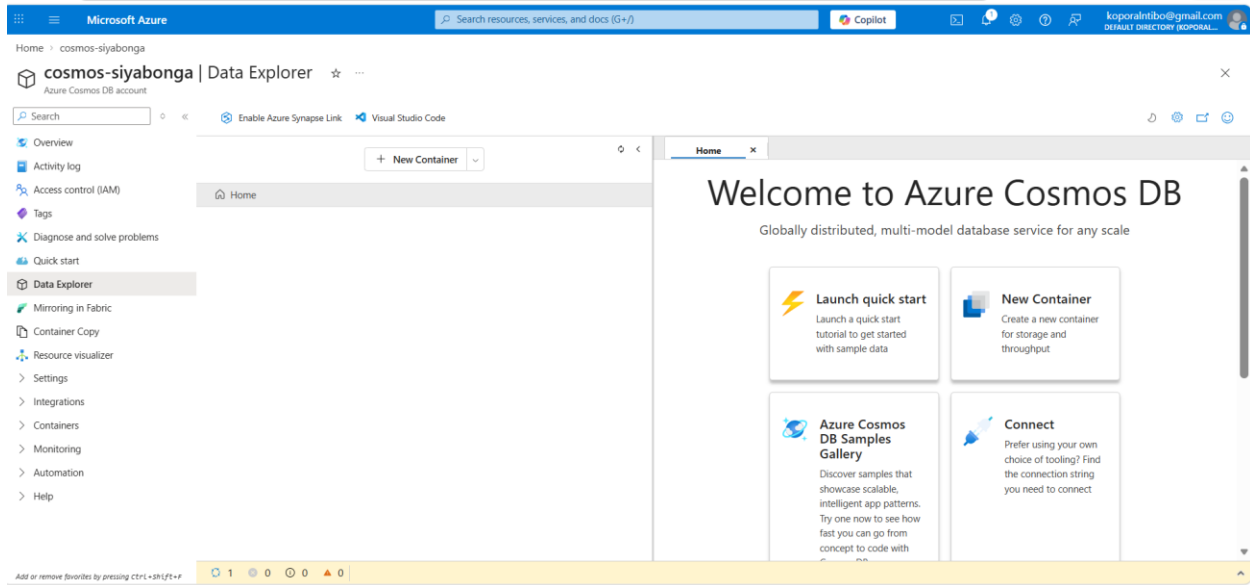
[Start learning today >](#)

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support. [Find an Azure expert >](#)

<https://portal.azure.com/#>

TASK 6: CREATE DATABASE AND CONTAINER



New ItemSaveDiscardUpload Item

HomeProducts.Ite... x

SELECT * FROM cType a query predicate (e.g., WHERE c.id='1'), or choose one from the drop down list, or leave empty to query all documents.

Apply Filter

id

...

/produc...

🔄

1

2

3

4

5

```
{
  "id": "1",
  "productName": "Laptop",
  "price": 15000
}
```

New ContainerX

☒ Create new☐ Use existing

ProductDB

☐ Share throughput across containers ⓘ

* Container id ⓘ

Products

* Partition key ⓘ

/productID

Add hierarchical partition key

* Container throughput (autoscale) ⓘ

☒ Autoscale☐ Manual

Your container throughput will automatically scale up to the maximum value you select, from a minimum of 10% of that value.

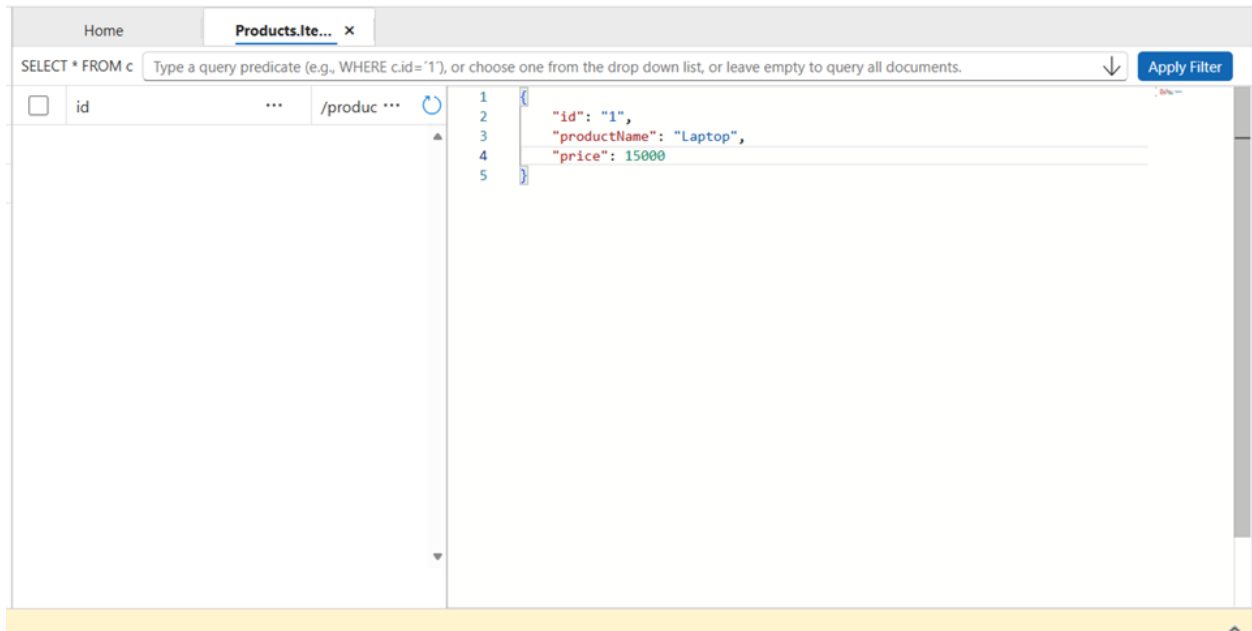
Minimum RU/s ⓘ

Maximum RU/s ⓘ

100x 10 =1000*

Task 7: Insert JSON Document

Inserting a JSON document is the standard way of working with NoSQL databases like Azure Cosmos DB, because JSON provides a flexible, schema-less format that fits perfectly with how NoSQL stores and queries data. Use the data provided to create a new item in the newly created container.



Home

Products.Ite... x

SELECT * FROM c

Type a query predicate (e.g., WHERE c.id='1'), or choose one from the drop down list, or leave empty to query all documents.

↓

Apply Filter

✓

id

...

/produc ...

↺

✓

1

1 {

2 "id": "1",

3 "productName": "Laptop",

4 "price": 15000,

5 "_rid": "-vIeAIHB-tMBAAAAAAAAA==",

6 "_self": "dbs/-vIeAA=/colls/-vIeAIHB-tM=/docs/-vIeAIHB-tMBAAAAAAAAA==/",

7 "_etag": "\"0b007861-0000-3000-0000-69c00ff00000\"",

8 "_attachments": "attachments/",

9 "_ts": 1774194672

10 }

Load more